

Double MSc: Quantitative Finance (ETH Zürich/UZH) + Applied Mathematics (HSE Moscow). Researching decision-making under uncertainty and market frictions in low SNR — transaction costs, forecast uncertainty, price impact, estimation error, and regime dynamics — to recover implementable strategies from theoretical ones. Prior research and industry experience span transaction costs, optimal execution, statistical arbitrage, volatility forecasting, and deep hedging.

WORKING PAPERS

- 2026 “**Shrinkage Estimation of Covariance Matrices: Economic Loss Minimization with Reinforcement Learning**” lead author, with Gianluca De Nard, Urban Ulrych — In preparation for submission
Economic loss regime-dependent shrinkage via VAR contextual bandits over RFF-encoded states; Bayes-optimal policy vs statistical surrogates. Closes one-third of the gap between data-driven methods and oracle.
- 2026 “**Strategy-Agnostic Cost-Aware Position Smoothing via Contextual Bandits**” lead author, with Urban Ulrych — In preparation for submission
Learns trade-speed from implicit return vs turnover trade-off via FPCA contextual bandits for exogenous weights as opposed to explicit Gârleanu–Pedersen objective. Improves gross-to-turnover ratio by 30–80 bps.

TALKS & CONFERENCE PRESENTATIONS

- Sep 2026 **Contributed Talk on RL Covariance Shrinkage paper (w/ De Nard, Ulrych)**, *Freddy Delbaen Conference (accepted)* (“From Analysis to Mathematical Finance and Insurance”) — ETH Zürich, 7–9 Sep 2026
- Jun 2026 **Poster on RL Covariance Shrinkage paper (w/ De Nard, Ulrych)**, *3rd ETH–HK–Imperial Joint Workshop on Quantitative Finance* — ETH Zürich, 22–25 Jun 2026 (invitation-only; admitted as MSc student)

EDUCATION

- 2024–now **ETH Zürich & University of Zürich** (Zürich, Switzerland) — D-MATH (ETH) & Dept. of Finance (UZH)
MSc in Quantitative Finance
Relevant courses: High-Dimensional Statistics with Prof. Bühlmann (6/6), Probabilistic AI with Prof. Krause (5.75/6), ML in Finance & Complex Systems (5.5/6), Quant AM and Systematic Investments (6/6), Quant PM with Python (6/6), Investments Quant Tools (6/6), AM: Advanced Investments (5.75/6).
Audited (no exam): BM & Stochastic Calculus, Theoretical Foundations of RL, Applied Stochastic Processes
Selected courses next semester: Mathematical Foundations for Finance, Dynamic Programming & Optimal Control, Advanced ML, Deep Learning, Time Series Analysis, Reinforcement Learning
Expected GPA 5.6 / 6.0
- 2022–2024 **NRU Higher School of Economics** (Moscow, Russia) — Faculty of Computer Science
MSc in Applied Mathematics & Informatics, Major: **Machine Learning and Data-Intensive Systems**
Relevant courses: RL (10/10, **top-1 of the course**), DL (10/10), NLP (10/10), Big Data (9/10), SE (9/10)
Thesis (Grade 10/10): “Hedging Derivatives Under Incomplete Markets with Deep Learning” (“**HSE Scientific Research Challenge 2024**” Awardee) — [GitHub](#).
GPA 8.9 / 10.0 (with Distinction)
- 2018–2022 **NRU Higher School of Economics** (Moscow, Russia) — International College of Economics and Finance University of London (under academic direction of London School of Economics)
Dual degree BSc in Economics and Finance
Thesis (Grade 10/10): “Expected Performance Optimization of Fixed Income Portfolio via Structured Products” (“**Best Thesis Class 2022**”, “**HSE Research Challenge 2022**” Awardee).
GPA 8.1 / 10.0

TEACHING EXPERIENCE

- Jan 2026–now **National Research University Higher School of Economics: Faculty of Computer Science**
Reinforcement Learning (320 students) — Co-Lecturer (DP, Q-Learning, TRPO/PPO, RL Finance), Head TA
- Aug 2023 **Python [FX market-maker simulator game](#) for MSc Derivatives Course** (Used by 120+ students)

OPEN-SOURCE SOFTWARE

- 2025–now “**Quant-PML**”: [Python library for ML Equity statistical arbitrage \(sklearn-like API\)](#) (pip install quant-pml)
- 2023–2024 “**Deep-Hedging**”: [Data-driven Deep Hedging / RL under TC-costs](#) (pip install deep-hedging)

RESEARCH AFFILIATIONS

Sep 2024– May 2026	University of Zürich Portfolio Management Program — <i>Qube RT Team</i> (Zürich) Highly selective (18 / 100+ applicants admitted, 7 graduated) 2-year hands-on research program; bi-weekly research meetings at QRT's Zürich; presented research every 3 weeks to UBS / QRT / UZH audience. Year 2: implemented live deploy of market-neutral equity stat arb strategies on QRT Q-Sec platform. Led strategies research and production implementation for the whole team.
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RESEARCH PROJECTS — *All topics self-proposed (own ideas / extensions of the cited papers)*

Apr 2026	PMP UZH: Ensembling Momentum Under Transaction Costs (<i>Grade To Be Issued Soon</i>) A data-driven ML approach to ensembling Momentum signals into a single confidence-aware strategy.
Mar 2026	PMP UZH: Contextual Bandits For Uncertainty-Aware Portfolios (<i>Grade To Be Issued Soon</i>) A new approach to Factor Portfolios construction that implements Bayesian Uncertainty No-Trade Region.
Dec 2025	PMP UZH: Volume Shocks and Overnight Returns (<i>Grade 6/6</i>) Improvement of Cartea et al. (2025) by Volume Prediction, based on Goyenko et al. (2024).
Dec 2025	DeepAR Distributional Forecasts for H-Step-Ahead Confidence Interval in Short Strangle Option Strategy (ETHZ Deep Learning — <i>Grade 5.5/6</i>) Predicting strangle strikes that yield a target exercise probability.
Jun 2025	Inverse RL & Behavioral Cloning For Covariance Shrinkage in Global Minimum Variance Portfolios (UZH Quant AM & SI — <i>Grade 6/6</i>) Improved De Nard & Kostović (2025) with Imitation Learning theory.
May 2025	Bayesian LSTM Ensemble For Short-Term Volatility Prediction With Orderbook Features (ETHZ ML For Finance and CS — <i>Grade 6/6</i>) LSTM and HAR regime-aware ensemble by Bayesian Uncertainty.
May 2025	PMP UZH: Term Paper — ML-Enhanced Fractional Momentum Selection and Timing (<i>Grade 5.5/6</i>) Improvement of Chitsiripanich et al. (2022) by Dynamic ML Optimization and Volatility Rebalancing.
Jan 2025	PMP UZH: High-Dim Sparse Additive Lasso and Uncertainty For Virtue of Complexity (<i>Grade 5.5/6</i>) Extension of Kelly et al. (2021) VoC framework by Gaussian Process with Timing by Bayesian Uncertainty.

PROFESSIONAL EXPERIENCE

Jul 2026–now	GSA Capital, Quantitative Research Intern (Deep Learning) — <i>Zug, Switzerland / London, United Kingdom</i> Raiffeisenbank JSC (part of RBI Group, Vienna) — <i>Moscow, remote from Zürich since Aug 2024</i>
2023–Jul 2026	Quantitative Investment Strategies — leading 9-person QR team, 12 strategies, \$815M AUM SVP & Product Owner (Nov 2025–), VP (Nov 2024–Oct 2025); Senior Officer (Jul 2023–Oct 2024) • FX Futures / Perps Stat Arb: OU parameters via DL state-space model; live as cost-aware EWMA policy • ML Multi-Signal Factor Strategies on Shanghai, Shenzhen, RU Exchanges (3 strats \$100M+ AUM) • Academic Beta Factors Cost-Aware Replication via Reinforcement Learning (\$60M+ AUM)
2022–2023	Product Manager, Prime Brokerage
2021–2022	Quantitative Analyst, Structured Products • Stat Arb for offshore vs onshore volatility and MOEX FX correlation trade in EURRUB & USDRUB
2020–2021	Sales Trader, Institutional Fixed Income, Sales&Trading
2019–2020	Aricapital AM — Intern, Fixed Income; Rosbank, Société Générale Group — Intern, Macro Research

EXTRACURRICULAR STUDIES

Summer School in Finance (Investment Management), WU Vienna — <i>Grade 1.0, 100%</i> (Vienna, August 2025)
SoFiE Summer School in Financial Machine Learning, Yale University (New Haven, July-August 2025)
AI in Finance Summer School, Nanyang Technological University (Singapore, July 2025)
The Experts: School of Quants — <i>Top-1 in Portfolio Optimization and Risk Management Exam</i> (2023-2024)

HACKATHONS

Dec 2024	ETH Probabilistic AI Course: Continuous Reinforcement Learning Competition (<i>Rank: 4/400+</i>).
Sep 2023	Winner of Open Hackathon in Quantitative Finance, hosted by The Experts: School of Quants

SKILLS & INTERESTS

Languages	English (proficient, IELTS 8.0/9.0), Russian (native)
Programming	Python (PyTorch, JAX, polars, gymnasium, hf-transformers), MATLAB, Java, C++, R, Golang, SQL
Interests	strategic card games (Preferans), fencing (national U18 medalist), football, poker, game design, skiing